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PISA 2022 Creative Thinking Assessment: Opportunities, Challenges, and Cautions

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ABSTRACT

The OECD's PISA program assesses 15-year-old students globally in key competencies every 3 years, providing influential data on education quality and spurring policy debates. In the latest cycle, the innovation domain focused on creative thinking, assessing over 140,000 students across 60+ countries, in the largest study of adolescent creativity to date. This innovation domain included a cognitive test covering multiple creative thinking processes (generating creative ideas, generating diverse ideas, and evaluating/improving ideas) and creativity domains (written expression, visual expression, social problem-solving, and scientific problem-solving), as well as an extensive survey on factors influencing creativity (such as openness, creative self-efficacy, or growth mindset). While this dataset offers unprecedented research opportunities due to its scale and international scope, challenges arise from its aggregated scoring and complex sampling design. Missteps in using this data in secondary analyses could lead to fragmented and inconsistent findings. This paper provides an overview of the PISA 2022 creative thinking assessment's framework, methods, and findings, highlighting both the potential and the caution needed for impactful creativity research.

1 | Introduction

Since 2000, the Organization for Economic Co-Operation and Development (OECD)'s Programme for International Student Assessment (PISA) has aimed to evaluate education systems worldwide by testing the skills and knowledge of 15-year-old students in participating countries/linguistic communities and economies. Every 3 years, a randomly selected representative sample of over 500,000 students is tested in reading, mathematics, and science, along with extensive self-report data designed to contextualize and help interpret the findings. The results are used to popularly "rank" countries on how their students match up on these core competencies. Reports with PISA results and analyses often serve as fodder for country-level policy debates and reforms. In addition to these competencies, PISA develops an innovative domain assessment for each cycle that targets a new and relevant 21st-century competence.

For the 2021 edition (relabeled 2022 due to delays from the COVID crisis, with findings and data eventually released in 2024), the innovation domain focused on creative thinking (CT; OECD 2024a). This domain assessed the creativity of over 140,000 teenagers across more than 60 participating OECD countries/economies, representing the largest and most international data collection on CT in history. It encompassed a cognitive battery of 32 CT items tapping into three thinking processes [generating creative ideas (GCI), generating diverse ideas (GDI), and evaluating/improving ideas (EII)] applied to four domains (written expression, visual expression, social problem-solving, and scientific problem-solving), as well as an extensive survey of

Both authors were members of the creative thinking expert group (CTEG) that contributed to shaping the competency model, developing items, and guiding the PISA 2022 CT Assessment progress from 2018 to 2023. We thank Natalie Foster and Mario Piacentini for their suggestions on a prior version of this manuscript. Baptiste Barbot is supported by the 2025–2027 Francqui Foundation Research Professor Mandate.

creativity enablers and resources (including creativity-relevant traits such as openness and creative self-efficacy; participation in creative activities; and school and familial support of—and opportunities for—creativity). While the recent release of the PISA CT assessment report (PISA 2022 Vol. 3, OECD 2024a) will undoubtedly fuel education policy debates on the importance of creativity in school, the public availability of PISA CT assessment data represents both a unique opportunity and a challenge for the field of creativity research.

It is an opportunity, as the size and richness of the dataset (such as its international scope) allow researchers to empirically test theoretical questions, explore a myriad of research possibilities, and apply sophisticated methodological and statistical techniques that are often not feasible given the smaller samples used in most creativity studies. Such approaches may lead to important new insights into adolescent CT that may be more generalizable than the smaller empirical studies that more typically comprise the creativity research literature.

However, there are also challenges. For one, PISA CT measured creativity across multiple domains and types of CT, yet only one "general" CT proficiency score—in fact, 10 alternate Plausible Values scores of this aggregate—was ultimately calculated and made public. These general proficiency scores were derived for all participants from countries enrolled in the 2021 innovation domain, using imputation techniques that estimate CT proficiency based on students' proficiency in all domains and other characteristics. With this method, respondents might be assigned a general CT proficiency score, even if they did not complete a single CT item; indeed, this was the case for about 70% of the sample.

The scope, size, and public nature of the project mean that many scholars and research teams will be tempted to dive into the dataset to publish, yet this is one case where the first will be hard-pressed to be the best. A multitude of studies, which will likely rely solely on that single score, may flood the field in an attempt to leverage the large dataset into a quick and easy publication. However, without precision and coordination, the ensuing glut of papers that will emerge—of widely varying quality—may lead to incoherent and inconsistent findings that result in more confusion than clarity on the nature of creativity. With collaboration, precision, and care, the PISA CT assessment could be a wonderful boon to the field of creativity scholarship. With chaos and competition, it could not only be a missed opportunity but an actual detriment to the striking progress that has happened over the last two decades.

Hence, we urge the field to not only coordinate research efforts but to develop methodological guidelines regarding secondary statistical analyses of this extensive dataset. Indeed, the PISA dataset is quite different from one that might emerge from a "typical" psychological or educational study. Its use of representative national samples means that there is stratified sampling and the need to weigh cases; the data, by its nature, is clustered. For many scholars in the field of creativity research, such large-scale assessments (LSAs) are new territory.

This paper aims to provide a firsthand overview and discussion of the PISA 2022 CT assessment, theoretical framework, methodology, and key findings. By doing so, it also seeks to provide an overview of the opportunities and challenges the PISA 2022 CT assessment data present for creativity research and education policy. Each of these broad topics will be discussed in greater length by the various contributors to the special issue, so the main goal here is to provide a general outline of these issues and highlight the most pressing conceptual and methodological questions one may have when discovering the PISA Vol. 3 reports, extensive methodological notes, and underlying data.

Correspondingly, we have multiple goals for this special issue. For one, we hope to initiate scientific discussion, debate, and stimulate coordination and potential collaboration between like-minded researchers. We will hear from a wide array of scholars representing many different disciplines, specialties, and approaches. We hope that the contributions that will follow will offer a cornucopia of theoretical, conceptual, methodological, statistical, and technical recommendations, best practices, and general suggestions that will help anyone who is considering utilizing the PISA data in future secondary analyses. We will conclude this special issue with an integrative conclusion that incorporates and synthesizes the collected wisdom and insights of our fellow scholars.

2 | The PISA Methodology: Standards, Strengths, and Concerns

Before diving into the PISA 2022 CT assessment, it is important to understand the general PISA goals, methodology, and related concerns that have been formulated in the literature. The PISA methodology has slightly evolved throughout the assessment cycles and is extensively described in an imposing technical report (OECD 2024b). In essence, PISA targets 15-year-old students attending the 7th grade or higher in both public and private schools. This age was chosen as it closely aligns with the end of compulsory education in most countries. The developmental window under focus is therefore extremely narrow (in the 2022 cycle data, ages ranged from 15.2 to 16.4 years old).

In terms of sampling, PISA uses a two-stage stratified sampling design. In the first stage, schools are sampled; in the second stage, students within those schools are sampled. This, together with an extensive quality control process (further presented below), ensures that the sample is representative of the entire population of 15-year-old students in each participating country. However, sampling and quality control alone cannot ensure that the sample captures the representativeness of the 15-year-old population in a given country. Hence, during the analytical stage, respondent weights are used to adjust for the fact that different students have different probabilities of being selected to complete the PISA survey (e.g., students in smaller schools have a higher probability of being selected than those in larger schools) and for nonresponse biases (if certain groups of students are less likely to participate, their responses are weighted more heavily to compensate for their underrepresentation). Ultimately, weights are used for calculating accurate estimates of population parameters such as average scores and proficiency levels and their associated standard errors and confidence intervals. Failing to use weights in secondary data analyses may result in biased estimates and associated standard errors, and ultimately, misleading conclusions (OECD 2009).

The cognitive assessment battery encompassed the three core domains of mathematics, reading, and science (as well as the CT innovation domain for participating countries) and included a combination of multiple-choice questions, closed- or short-response questions, and open-constructed response questions. Assessment items were distributed to students through multiple test forms designed for an approximately two-hour testing session. Each form contained a balanced mix of items from the main domains and was randomly assigned to students to ensure fairness and representativeness. In practice, this means that no single participating student received the entirety of items for a given domain. In other words, data missingness is the rule, not the exception. This has, of course, critical implications as to how both missingness "by-design" and missingness due to nonresponse should be handled in secondary analyses (e.g., Rutkowski et al. 2010).

In addition to the cognitive assessments, PISA administered various "context" questionnaires to students (as well as school principals, parents, and teachers) following the same distribution and complex sampling design described for the cognitive assessment. These questionnaires gathered contextual information about students' backgrounds, attitudes, and learning environments. Some were specifically related to creativity, and others were not.

The PISA survey was typically administered in a controlled environment, often within a classroom setting or a computer lab. PISA is now exclusively administered as a computer-based assessment. While the shift to digital testing has allowed for the inclusion of more interactive and adaptive assessment tasks in PISA, it may also further raise concerns about the digital divide and the complex effects digital literacy and access have on performance (e.g., Tan and Hew 2019). Each student completed the assessment individually (but in a group setting) under the supervision of trained personnel to ensure standardization so that any technical issues that might arise could be addressed.

At all stages of PISA's implementation, a comprehensive data quality control process was conducted to ensure the accuracy and reliability of results. This included a careful review of the translation and adaptation of survey materials to ensure they met PISA's technical standards and were culturally appropriate; field trials that tested and refined the instruments and procedures before the launch of the actual data collection; visits to data collection sites by different instances to monitor the implementation of the survey at the international level and ensure consistency and adherence to quality standards; close training of test administrators; and a careful review of procedures for test administration, training coders, verifying coding procedures, and managing the data entry and submission processes.

Despite the rigor and scale of the PISA survey, concerns with the general PISA initiative, content, methodology, or policy implications abound in the literature (e.g., Sjøberg and Jenkins 2022). An extensive review of classic criticisms of PISA (and other LSAs) is out of scope here. However, we thought that a highlight of key concerns could be outlined as a general context to understand what PISA is or is not. Such concerns can also serve as a warning for readers unfamiliar with PISA about existing limitations that have already been the subject of discussion, with the goal of helping them better contextualize and understand the nuances of the PISA Vol. 3 findings as well as any future results stemming from secondary data analyses.

Among criticisms, it is often said that PISA's assessments may be biased toward Western, industrialized educational models, which can disadvantage students from other cultural or educational contexts (Zhao 2016). Since the test content and interpretation are standardized across diverse countries, the assumption that such disparate education systems should perform equally on the same metrics has been seen as problematic, even extending to various local contexts within the same system (Hopmann 2008).

In terms of statistical approaches, some researchers have questioned PISA's use of plausible values (i.e., statistical method used to estimate students' ability), specifically their reliance on Rasch modeling (Baird et al. 2011; Goldstein 2004). These critics argue that such methods make strong statistical assumptions and obscure the actual abilities of individual students, thereby potentially creating results that are not as reliable as they seem.

Another criticism concerns the "one-size-fits-all" approach to education. The focus on core competencies and global accountability (i.e., country ranking) has been seen as promoting a narrow view of education (Sellar et al. 2017). The reliance on PISA rankings oversimplifies complex educational issues and can lead to policy reforms that are solely aimed at improving scores, rather than addressing deeper educational challenges (Breakspear 2012; Komatsu and Rappleye 2017) and emphasizing a well-rounded education (Meyer and Benavot 2013; Sjøberg 2015). The inclusion of CT in PISA 2022 has helped address the first concern around PISA's narrow focus on core competencies. However, the recurring issue of potentially oversimplifying complex educational issues nonetheless does seem to also apply to PISA 2022's CT assessment.

3 | The PISA CT Assessment: Context, Competency Model, and Further Concerns

It has not escaped the attention of the OECD that creativity is a crucial skill for youth in the 21st century. Before incorporating CT as part of the last PISA cycle, several OECD publications (e.g., Lucas et al. 2013) and research initiatives from OECD's Centre for Educational Research and Innovation (CERI), have placed CT at the forefront of educational innovation and policy development, emphasizing its critical role in preparing students for the demands of the 21st century. Bringing CT into PISA, a program geared towards core academic competencies of mathematics, reading, and science, may appear to be a significant leap, and surely a bold ambition for the OECD. However, it remains consistent with PISA's broader focus on assessing students' ability to *apply* their knowledge to real-world situations. Their aim is to measure not only core academic competencies but also students' preparedness for the problem-solving and decision-making needed in the modern world.

Hence, by focusing on CT, PISA aimed to highlight the importance of preparing students for the challenges and opportunities of the future by fostering creativity in education systems worldwide. It was also, perhaps, a way to address a recurring criticism of PISA's narrow focus, including with regard to the core competencies under investigation (e.g., Sjøberg 2015; Zhao 2016). As stated in PISA's Creative Thinking Framework (OECD 2019), "Developing an international assessment of creative thinking can encourage positive changes in education policies and pedagogies. The PISA 2021 Creative Thinking assessment will provide policymakers with valid, reliable and actionable measurement tools that will help them to make evidence-based decisions. The results will also encourage a wider societal debate on both the importance and methods of supporting this crucial competence through education" (p. 5). However, it is important to note that this ambitious objective was quickly short circuited by several countries—including the United States and the United Kingdom—opting not to participate in the CT innovation domain during the 2022 PISA cycle. Reasons for opting out included varying educational priorities or concerns about how CT would be measured and implemented on a large scale.

So how is creativity measured in PISA? The PISA CT competency model (OECD 2019, 2024a) includes three types of CT [*GDI, GCI*, and *evaluating and improving ideas (EII)*] across four domains (*written expression, visual expression, scientific problem-solving*, and *social problem-solving*). Developed in consultation with creativity experts, the model reflected, to some degree, current perspectives that include both multiple CT processes (e.g., Amabile and Pratt 2016; Boldt and Kaufman in press; Lubart et al. 2011; Wigert et al. 2024) and a balance between domain specificity and generality (Baer and Kaufman 2017; Barbot et al. 2016; Chen et al. 2020; Lubart et al. 2011; Qian et al. 2019). The data collection proceeded to follow this competency model, resulting in a large array of creativity tasks (ultimately, 32 items were retained). However, although the PISA Vol. 3 report (Chapter 4) does present a series of findings that distinguish the various thinking processes and domains, the decision was ultimately made to merge all of these different tasks that measured these different thinking processes and domains into one large, aggregated global score (CT proficiency score). This choice may have been consistent with the PISA "metric" but it is much more akin to decades-old approaches that assumed a domain-general perspective than to current ways of thinking about creativity (Glăveanu and Kaufman 2019; Kaufman and Glăveanu 2019).

It is an open question whether the thinking processes and domains chosen were the right ones. There are many others that could have been selected; for example, domains that regularly appear on domain-specific self-report creativity checklist measures include acting, singing, dancing, composing, humor, cooking, sports, invention technology, and many others (Carson et al. 2005; Diedrich et al. 2018; Kaufman 2012). However, the selections appear reasonable considering the usual domains of creative activity among 15 years old as well as the practical limitations of the testing scenario. The core domains of writing, art, science, and social/everyday creativity are also represented on most domain-specific self-report creativity checklist measures (e.g., Carson et al. 2005; Diedrich et al. 2018; Kaufman 2012). Furthermore, one can also question the operationalization of the three key thinking processes addressed in the competency model. For instance, most tasks measuring PISA's *EII* consist of generating an original response that goes beyond a provided idea. Although this design choice was made to limit interdependence between items and optimize standardization, one can debate or question the actual involvement of evaluative processes in such tasks.

However, what we see as a more central question is whether the choice to use a single, composite score was justified with regard to the observed structure of the data. This is not only a "psychometric" question. This issue concerns an important assumption regarding the nature of creativity, leading to critical consequences in the officially presented results in the PISA Vol. 3 report and future findings on CT using the PISA CT data. Many questions logically stem from this initial assumption. First, to what extent was item selection devised to converge towards fitting a unidimensional scale in the test development process? What does this single composite score actually represent? Is it a good summary of teens' ability to think creatively? How were the plausible values estimated for students who did not complete any CT items—which were about 70% of participants? Since plausible values are calculated using all available data, might this explain the high correlations found between proficiency in CT and proficiency in the other PISA core domains (an intriguing finding that we discuss below)? Should creativity scholars continue to use this general proficiency score in future secondary analyses of PISA data to represent the sole indicator of CT abilities? These questions will be extensively discussed throughout this special issue.

There are certainly many possible reasons why, despite the multidimensional nature of the CT competency model and assessment, the results were primarily reported as a general proficiency score. Most likely, this decision was made for ease of interpretation and to allow comparability across different countries and cultures. However, it raises questions about the adequacy of a single aggregated score in capturing the full scope of CT abilities (that are generally rather independent). Although built upon a completely different philosophy and methodology, this index ultimately resembled other indices such as the Global Innovation Index, Global Creativity Index, or the European Innovation Scoreboard. In the context of PISA, ranking countries on a single general index of CT may be useful for educational policy debates. It is less useful in understanding creativity as a multifaceted human ability.

With this in mind, it is also important to better contextualize the PISA 2022 CT assessment data. For one, while PISA is highly standardized, one should remember that its particular testing context (e.g., often classroom settings with adult supervision) may have a nonnegligible impact on creative performance (e.g., Rubenstein et al. 2018; Yuan and Zhou 2008). It is also important to note that some of the testings occurred during the COVID-19 pandemic when many countries were in lockdown mode. The impact of the pandemic and lockdown on student creativity is complex; many people self-reported being more creative during the lockdown (e.g., Lopez-Persem et al. 2022; Mercier et al. 2021). Creative students were more likely to experience depression and anxiety during this time (Kerr et al. 2021), yet creativity also represented an enjoyable component of schoolwork (Patston et al. 2021).

An additional point to consider is the specific age range tested by PISA (15-years old) which represents a unique period in terms of CT development. It notably falls during the 7th grade creativity "slump," or at least a possible "slump recovery" period (Said-Metwaly et al. 2020). It also corresponds to a time of specialization of creativity towards creative activities that adolescents invest in more closely (Barbot and Rogh 2020; Barbot and Tinio 2015). Therefore, PISA findings, as well as future findings from secondary data analyses, should situate results in light of this particular developmental context. Such results may be unlikely to generalize to creativity "in general" or to the whole lifespan, and it is essential that scholars take this larger context into account.

Moving on to the self-report context data from the students, parents, teachers, and principals, a wide array of constructs was measured, from creative self-efficacy to openness to access to resources that might support creative expression. However, a brief examination of the scoring map suggests that these constructs should be further examined in terms of construct or content validity.

To conclude, we have only raised issues that struck us as particularly salient or noteworthy. There are many additional such questions that encompass both creativity theory and past findings (as well as general hierarchies of scientific decisions) that can and should be raised and addressed for numerous aspects of the PISA CT report.

4 | Findings From PISA 2022 CT Assessment and Policy Implications

There are several unexpected findings in the PISA Vol. 3 report when one considers the large scope of existing theory and research in the field of creativity. Many of the results that stand out as being unusual or inconsistent with creativity scholarship may not be so independent from PISA's choice to use a single composite indicator to conceptualize a wide range of CT abilities and domains. Yet, even assuming CT can be measured unidimensionally, some key findings in the PISA Vol. 3 report appear strikingly inconsistent with what we know as a field. Here, we will only provide a cursory overview of such findings, as they will be further discussed in this special issue.

First, the PISA Vol. 3 report suggests a rather large and consistent (i.e., observed across almost all participating countries/economies) gender gap in CT. Girls outperform boys across all types of CT processes and domains (though the gap is more marked in visual, verbal, and social problem-solving than in scientific problem-solving, with slightly different patterns across countries). Although recent studies and meta-analyses have outlined rather small differences in mean performance in CT tasks with small female advantage if any (Abdulla Alabbasi et al. 2022; Taylor et al. 2024; Taylor and Barbot 2021), the differences appear sizable in PISA Vol 3. report, with the subtitle "*Girls are considerably stronger creative thinkers than boys*" appearing in OECD's report highlight. Remarkably, compared to boys, girls are indeed also overrepresented in the Level 5 top-performers group in CT (on average 31% vs. 23%). This means that a larger proportion of girls than boys have achieved the highest proficiency level in the PISA 2022 CT assessment. This finding is also inconsistent with recent meta-analyses (Taylor et al. 2024) and studies (Taylor and Barbot 2021) testing the Greater Male Variability Hypothesis in CT, which instead suggest a trivial (Taylor et al. 2024) to small (Abdulla Alabbasi et al. 2022) variability advantage for males. Accordingly, these meta-analytic findings suggest that, if anything, a minimal overrepresentation of male, rather than female, students should be observed in the top-performers group of PISA CT. Beyond some explanations outlined in PISA Vol 3. report (e.g., girls' reported attitudes, beliefs, and overall greater engagement with the PISA CT test could account for this), what can explain these discrepancies?

Another striking finding is that the PISA Vol 3. report suggests a rather strong overlap between performance in CT and reading, mathematics, and science. This finding exceeds expectations of effect sizes suggested by studies on the link between academic performance and CT (Gajda et al. 2017). Indeed, PISA data revealed correlations of CT reaching 0.67 with mathematics, 0.66 with reading, and 0.66 with science, which is well above the 0.22 average correlation estimate across 120 studies (Gajda et al. 2017). The contrast is even more striking when translating these correlations into shared variance, revealing up to 45% of shared variance between CT and the other PISA domains compared to an average 5% shared variance in creativity-academic achievement studies. However, one should keep in mind that PISA science and mathematics have a 0.87 correlation, and reading has a 0.80 correlation to both mathematics and science. Given this context of high interdomain correlations in PISA, it is somewhat reasonable to agree that PISA's "*...creative thinking assessment measures a different subset of skills with respect to those measured in the mathematics, reading and science assessments*" (p. 83). It is similarly possible to some extent to agree with the PISA Vol. 3 report's conclusion that creativity can flourish independently of high academic performance. With that in mind, one can still wonder what this 45% shared variance between CT and core academic domains actually represents. Is CT more involved in those academic domains than previously thought? Are CT items in PISA relying too much on basic academic competencies? Or are there perhaps methodological and statistical explanations for this finding? For example, could the method used to estimate plausible values in CT be somewhat "contaminated" by students' performance in other domains? This concern arises because CT proficiency scores are estimated for a large portion of students who did not complete a single CT item in PISA 2022.

Finally, another surprising finding is the negative correlation between CT performance and participation in creative activities. Specifically, the PISA 2022 Vol. 3 report highlights a nuanced relationship, whereby frequent or irregular participation in creative activities, including arts, music, or creative writing, is negatively correlated with performance. The report invokes a self-selection bias as a possible explanation of this finding (e.g., how students who participate in creative activities may already differ in meaningful ways from those who do not). This seems contradictory with the broader assertion that creative engagement enhances creativity: "*the importance of creative participation in both formal and informal settings for enhancing students' creative thinking abilities. Schools that integrate creative tasks and provide opportunities for students to express their ideas, combined with extracurricular engagement,* *help students achieve better creative outcomes*" (PISA 2022 Vol. 3 report). This conclusion may possibly be valid with the caveat that creative participation needs to be balanced and meaningful to truly enhance CT. Overparticipation without depth, or very sporadic participation in many different areas, may potentially hinder creativity development given that such students' creative engagement lacks either consistency or motivation. This is potentially in line with theory and research that argues that frequency of creative participation alone does not guarantee improved creativity outcomes and that the development of creativity mainly requires engagement and opportunities to experiment (e.g., Beghetto and Kaufman 2014; Beghetto et al. 2014). However, given that creative activities, creative opportunities, and creative performance are much more likely to go hand in hand (e.g., Asquith et al. 2024; Diedrich et al. 2018; Eschleman et al. 2014), we hesitate to immediately accept it without much deeper further exploration. If subsequent investigations with the PISA CT dataset reinforce the report's conclusions, then there is much potential for future research that may consider not only how often students engage in creative activities, but also the way in which they do so.

Taken all together, the pattern of findings in the PISA report offers much beyond the popular country rankings that often trigger education policy debates and reforms. There are many potential implications for educational policy and teaching practices that may improve student outcomes in terms of CT. However, such implications may be premature at this stage, as the unexpected nature of the findings outlined above may require further investigation. This returns us to our central question, which can be summarized as follows: Have we really measured CT in PISA 2022? Hence, before addressing the education policy and educational practice implications, it appears essential to first initiate a thorough scientific debate and further investigation of the PISA 2022 CT assessment data.

5 | Conclusions: Opportunities and Challenges for Creativity Research

Despite the limitations of the PISA 2022 CT assessment as outlined in this paper, PISA should be commended for conducting the largest data collection on CT to date. The public dataset offers a potentially priceless and long-lasting gift to our research community. How the field utilizes this extensive resource will have critical implications for the future of creativity research and education policy. To fully leverage this opportunity, researchers must apply rigorous methodological standards and careful analyses. If our field can work together and follow best practices, it could potentially yield unprecedented, generalizable insights that could shape our understanding of creativity for decades to come. As outlined in the introduction, the opportunities presented by the PISA CT dataset are indeed manifold, including a better understanding of the nature of adolescent CT, its educational enablers and barriers (from education system-level to classroom-level variables), related gender differences that may potentially lead to a gender gap in creative achievements, and the contribution of CT to a myriad of other psychological constructs addressed in the last PISA cycle.

This first snapshot of PISA 2022 CT assessment conceptualization, methodology, and findings should prompt future rigorous secondary analyses of PISA data, which is far from a straightforward endeavor. In other words, we contend that the PISA Vol 3. report cannot currently be accepted as a sufficient basis for scientific insights and policymakers to "make evidence-based decisions" for educational reforms: a close scientific reexamination is necessary. To do so, the highest standards of scientific excellence in terms of both transparency and scientific integrity should be implemented. However, this brief overview of the PISA methodology and the extent of criticism brings about numerous methodological questions in view of secondary data analyses. To name a few questions that we hope scholars will consider in this special issue and related work: How should researchers handle missing data? Can plausible values be predicted at the item level, and what is the best way to do so? Should researchers derive new participants' weights if they are using a subsample of PISA data? Should process data (e.g., response time) be factored in to provide more granularity in estimating item-level performance (Barbot 2018) as it was successfully done with other PISA core domains (Han et al. 2019; Liu et al. 2018)? What is the best way to leverage PISA data by combining them with other, publicly available datasets? What are the best analytical choices when dealing with aggregated (country-level) data? Should researchers weigh (selected) countries by population size?

This special issue was designed to gather together a multitude of experts to provide key recommendations and guidelines for scholars who want to tackle the PISA CT dataset. Some of these pieces may simply offer best practices for any project involving LSA data, covering such topics as preregistering planned analyses, choosing the proper unit of analysis, paying careful attention to survey design, how to handle patterns of missing data, and considering issues related to sample representativity and weighting. Others may be specific to this particular dataset, such as making sure that the data being analyzed is consistent with the research questions being asked or highlighting conceptual or theoretical issues that should be considered.

Perhaps most saliently, we argue that there is a strong need to question the construct validity of the PISA 2022 CT assessment scores (and context questionnaire indicators). Although we are quick to challenge the usefulness and appropriateness of the unidimensional model of the cognitive battery, we must also point to other ways that this dataset can and should be considered and analyzed. Given this, we support ongoing and future efforts aimed at developing and testing alternative scoring (and measurement) models of PISA CT items, as opposed to relying on the current general proficiency scores and self-report-based indexes. Such efforts represent, in sum, the cornerstone of the success of any future project employing the PISA CT data. The follow-up question, then, must become whether more accurate, nuanced, and focused CT scores will yield results sufficiently consistent with the main findings reported in the PISA 2022 Volume 3 report, including country rankings?

At this stage, we therefore are more focused on asking questions than offering responses. It is our hope that the insights from the contributions to this issue will stimulate scientific debate and enable a general coordination of research efforts using this data. Further, it is our hope that the contributions in this special issue will provide solutions and help derive clear guidelines for future analyses. We hope that this issue can offer a roadmap that paves the way for years of creative research using the PISA 2022 data. If the wave of empirical reports that will come out from creativity scholars are done with this type of care and precision, then this work could have significant implications for creativity theory and educational policies.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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